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Magnetostratigraphic correlation of the Oxfordian–Kimmeridgian boundary

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ABSTRACT

A magnetic polarity pattern for Boreal and Sub-Boreal ammonite zones of the Upper Oxfordian to Lower Kimmeridgian was established and confirmed in four British sections, including the proposed Global Boundary Stratotype Section and Point (GSSP) on the Isle of Skye (Scotland) to define the base of the international Kimmeridgian Stage. A coeval pattern for Sub-Mediterranean ammonite zones was compiled from seven sections in Poland, one German section and multi-section composites from France and Spain. The mean paleopole for the European Craton (excluding Spain) at the Oxfordian–Kimmeridgian boundary is 74.2°N, 181.3°E (A_{95} = 3.8°). The common magnetic polarity scale enables inter-correlation of ammonite subzones among these three faunal provinces and to the marine magnetic-anomaly M-Sequence. The proposed GSSP at the base of the *Pictonia baylei* Zone is near the base of an extended interval dominated by reversed polarity, which is interpreted to be Chron M26r. This GSSP level projects to the lower to middle part of the *Epipeltoceras bimammatum* Subzone, which is the middle subzone of this *E. bimammatum* Zone in the Sub-Mediterranean standard zonation. In contrast, the traditional placement of the Oxfordian–Kimmeridgian boundary in that Sub-Mediterranean standard zonation (base of *Sutneria platynota* Zone) is at the base of Chron M25r, or nearly 1 million years younger.

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1. Introduction

Ammonites and other marine fauna evolved during the Late Jurassic within separate climatic provinces in quasi-isolation. Each of these provinces has a high-resolution zonation based on ammonites, but it has been difficult to achieve an unequivocal correlation among these zonations except for rare intervals. This provincialism hindered the establishment of a Global boundary Stratotype Section and Point (GSSP) for the international Oxfordian–Kimmeridgian boundary, because such a GSSP should provide the means for precise global correlation. Traditionally, the base of the Kimmeridgian in the Boreal faunal province has been placed at the base of the ammonite *Amoeboceras bauhini* Zone, and in the Sub-Boreal province at the base of the *Pictonia baylei* Zone. However, in the Sub-Mediterranean province, the Kimmeridgian has a quite different definition and span, and traditionally began with the base of the ammonite *Sutneria platynota* Zone. Quantifying the offset between these competing regional definitions and an agreement on a standardized international Kimmeridgian has been a challenge.

A proposed international definition of the base of the Kimmeridgian within the British section in Flodigarry, at Staffin Bay, Isle of Skye (Wierzbowski et al., 2006; Matyja et al., 2006) was accepted in January, 2007, by over 70% of the Kimmeridgian Working Group of the Jurassic Subcommission of the International Commission on Stratigraphy (ICS). However, before a GSSP can be established, it is necessary to resolve how this level can be confidently correlated with other sections in the world. Therefore, in addition to biostratigraphy from unique evolutionary trends within multiple taxa groups, it is desirable to identify less-unique, but more global, non-biological stratigraphic tools, such as stable isotope excursions, major eustatic sea-level fluctuations or distinctive magnetic reversal horizons.

Magnetostratigraphy, combined with biostratigraphic constraints, can be a powerful tool for high-resolution correlation of stratigraphic sections. The M-sequence of marine magnetic anomalies spanning the Late Jurassic provides a reference polarity pattern and numerical time

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scale derived from seafloor spreading models (reviewed by Ogg and Smith in Gradstein et al., 2004). The presumed Kimmeridgian through Tithonian portion of this M-sequence (chrons M25r through M19n) is dominated by relatively long-duration polarity chrons. Magnetostratigraphic sections in condensed pelagic strata of southern Spain had indicated that the traditional placement of the "Oxfordian–Kimmeridgian boundary" within the Sub-Mediterranean faunal province was near polarity zone M25r (Ogg et al., 1984), but this level was suspected to be significantly younger than the classic "Oxfordian–Kimmeridgian boundary" of the Boreal faunal province. Marine magnetic anomalies older than Chron M25 are both closer spaced and lower in amplitude. Deep-tow magnetometer surveys in the Pacific Ocean have established a tentative polarity model extending into the Bathonian (Sager et al., 1998; Tivey et al., 2006; Tominaga et al., 2008), but this model remains uncalibrated owing to the lack of a complete and verified magnetic polarity time scale for the Bajocian through Oxfordian. Particularly vexing is the interval between anomalies M25r and M27r that has been interpreted from less precise sea-surface magnetic surveys and not yet verified by deep-tow magnetometer investigations.

We have utilized magnetostratigraphy based on over 900 paleomagnetic samples to establish an inter-regional correlation of the ammonite zones/subzones for Upper Oxfordian and Lower Kimmeridgian stages based on combination of the bio- and magnetostratigraphic results from the Boreal and Sub-Boreal temperate faunal provinces of Britain, and the Tethyan tropical faunal realm of Poland, France, Spain and Germany. Rare episodes of Boreal-fauna (cool) incursions into the Tethyan (warm) basins of Poland and Germany enable only a few inter-regional correlations limited to narrow stratigraphic intervals or horizons (e.g., Schweigert and Callomon, 1997; Matyja and Wierzbowski, 1997, 2002, 2006a,b; Matyja et al., 2006). Our correlation of magnetostratigraphies from these regions enables assignment of several precise horizons of inter-regional correlation, and provides a reference polarity pattern for the poorly-resolved marine magnetic anomalies between M25r and M27r. The proposed GSSP for the Oxfordian–Kimmeridgian boundary on the Isle of Skye (Wierzbowski et al., 2006; Matyja et al., 2006) nearly coincides with a distinctive polarity reversal that we correlate to the base of Chron M26r. The correlation to the M-Sequence chrons quantifies the approximate age, offset and duration of ammonite subzones and other datums among the different faunal provinces.

2. Magnetostratigraphic sections

Paleomagnetic minicores were drilled in most of the sections with an average spacing of 0.5 to 1 m. Denser samplings were near stage boundaries, within condensed intervals and close to hiatuses as determined by sequence stratigraphic analysis.

2.1. Boreal and Sub-Boreal sections from Britain

The Flodigarry section located at the northern end of Staffin Bay is the leading candidate of the Kimmeridgian Working Group of the Jurassic Subcommission of the ICS for the basal Kimmeridgian GSSP (Wierzbowski et al., 2006; Matyja et al., 2006). This section contains both Boreal and Sub-Boreal ammonites, and the GSSP would presumably coincide with the approximately synchronous bases of the Boreal *Amoeboceras bauhini* Zone and Sub-Boreal *Pictonia baylei* Zone. However, there are two competing concepts for the base of the *P. baylei* ammonite zone: (1) the base of the *Pictonia flodigarriensis* horizon (Matyja et al., 2006; Wierzbowski et al., 2006; see also Fig. 4 herein), or (2) the base of the overlying *Pictonia densicostata* horizon (Schweigert and Callomon, 1997).

We sampled the proposed GSSP Flodigarry section on the Isle of Skye during two independent field campaigns. We also collected samples from the reference sections for Sub-Boreal biostratigraphy of Upper Oxfordian and Lower Kimmeridgian at South Ferriby, Ringstead Bay and Black Head (Fig. 1; Table 1, with detailed stratigraphy provided in the supplementary figures).

Oxfordian marine clay-rich strata along the NE coast of the Isle of Skye at Staffin Bay, Scotland, precisely described by Morton and Hudson (1995), are candidates for the global boundary stratotypes of both the Callovian–Oxfordian and Oxfordian–Kimmeridgian boundaries (Page, 2004; Matyja et al., 2006; Wierzbowski et al., 2006). These mudrocks are exposed in the inter-tidal zone in a series of small fault blocks that dip between 30° and 90° to the west. They contain an abundant ammonite fauna and provide the standard section for zones within the Sub-Boreal–Boreal provinces (Sykes and Callomon, 1979; Wright, 1989; Cox, 1990; Matyja et al., 2006). At the proposed GSSP section of Flodigarry, the leading candidate level for the Oxfordian–Kimmeridgian boundary at about 1.30 m below Bed 36 corresponds to the *Pictonia flodigarriensis* horizon — the first occurrence of Boreal *Amoeboceras praebauhini* and *Pictonia* with *Prorasenia* replacing *Ringsteadia* and *Microbiplices* Sub-Boreal ammonites (Matyja et al., 2006; Wierzbowski et al., 2006). This level is dated by the Rhenium–Osmium method in the Flodigarry section as 154.1 ± 2.2 Ma (Selby, 2007). The fault blocks are overlain and intruded by Paleocene basaltic lavas and dykes associated with the North Atlantic Igneous Province. The larger intrusions alter the chemistry and magnetization of immediately adjacent sediments (e.g., McClelland Brown, 1981, see Table 1), but the thermal effects have not reset the magnetization of Staffin exposures more than two dyke-widths from the dyke edge (McClelland Brown, pers. comm., 1997). Most of these intrusions are reversely magnetized (reviewed in Dagley et al., 1990), and we similarly observe a residual secondary reversed overprint in some of our mudstone samples. Two independent suites from the GSSP section

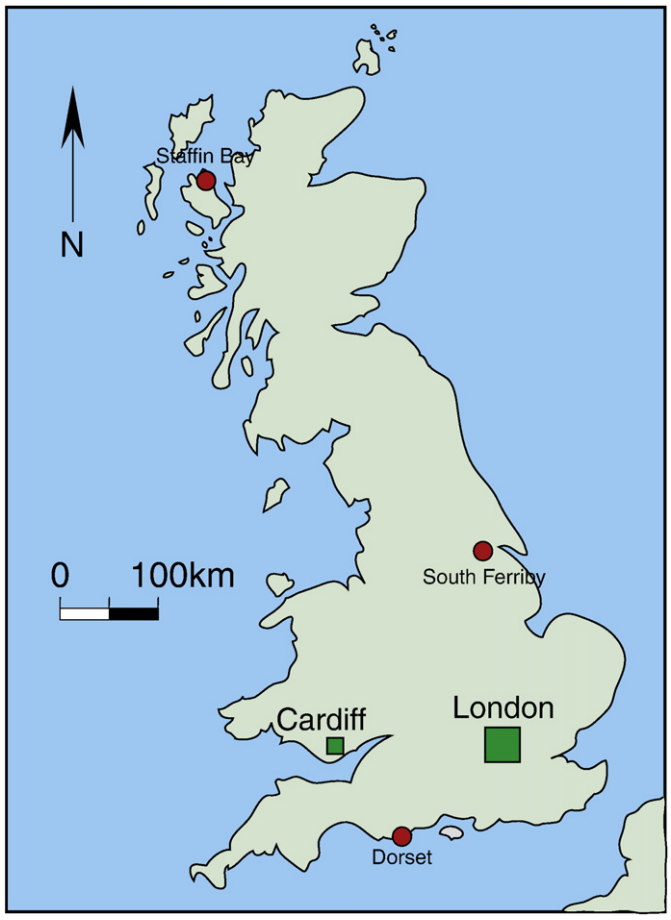


Fig. 1. Location of British sections in Boreal- and Sub-Boreal faunal provinces.

Table 1
Boreal, Sub-Boreal and Sub-Mediterranean magnetostratigraphic sections — location, position, stratigraphic interval, orientation, number of samples and year of sampling. [Cited stratigraphic references in are tabulated in Supplementary data to this article.]

Section	Location	Position		Biostratigraphic span of sampling	Average bedding		Analyzed samples	Year of sampling	References	
		Lat [°]	Lon [°]		Dip [°]	Dip Dir [°]			Lithostrat	Biostrat
Staffin Bay	**Isle of Skye, UK**	57°40'N	6°15'W	**Kitchini - Regulare**	**63**	**251**	**48**	**1995, 1996, 2002**	McClelland Brown 1981; Thrasher 1992; Dagley etal., 1990; Bishop and Abbott, 1995; Morton and Hudson, 1995	Wright 1989; Matyja et al., 2006; Wierzbowski et al., 2006;
South Ferriby	**Yorkshire, UK**	54°42'N	0°30'W	**Cymodoce - Pseudocordata**	**2**	**180**	**27**	**1995**	Cox 2001; Wright 2003	Birkelund and Callomon, 1985; Ahmed 1987; Wignall 1990; Schweigert and Callomon, 1997; Cox 2001
Ringstead Bay	**Dorset, UK**	50°36'N	2°24'W	**Cymodoce - Baylei**	**8**	**130**	**24**	**1995**	Coe 1992, 1995; Wright 2003	Cox & Gallois, 1981; Wright 2003
Black Head	**Dorset, UK**	50°36'N	2°24'W	**Pseudocordata - Pumilus**	**46**	**11**	**18**	**1995**	Coe 1992, 1995; Wright 1997, 2003	Wright 1986a, 1997a, 2003
Lisowice	**Polish Jura Chain**	51°07'N	18°50'E	**Planula**	**3**	**117**	**25**	**2007**	Matyja and Wierzbowski, 1997, 2006b	Matyja and Wierzbowski, 1997, 2006b
Lisowice2	**Polish Jura Chain**	51°06'N	18°50'E	**Planula - Bimammatum**	**3**	**360**	**26**	**2007**	Matyja and Wierzbowski, 1997, 2006b	Matyja and Wierzbowski, 1997, 2006b
Raciszyn2	**Polish Jura Chain**	51°07'N	18°51'E	**Bimammatum**	**3**	**360**	**20**	**2007**	Matyja and Wierzbowski, 1997, 2006b	Matyja and Wierzbowski, 1997, 2006b
Raciszyn	**Polish Jura Chain**	51°06'N	18°52'E	**Bimammatum**	**19**	**7**	**27**	**1992**	Wierzbowski 1992, Matyja and Wierzbowski, 1997, 2006b	Matyja and Wierzbowski, 1997, 2006b
Bobrowniki2	**Polish Jura Chain**	51°06'N	18°47'E	**Bimammatum**	**4**	**292**	**35**	**2007**		
Bobrowniki	**Polish Jura Chain**	51°06'N	18°47'E	**Bimammatum**	**6**	**337**	**36**	**2007**		
Syborowa Gra	**Polish Jura Chain**	50°17'N	19°34'E	**Bimammatum - Bifurcatus**	**12**	**115**	**34**	**1992**	Matyja and Wierzbowski, 1992, 2006a	Matyja and Wierzbowski, 1992, 2006a
Plettenberg	**Swabian Alb, Germany**	48°13'N	8°49'E	**Planula - Bimammatum**	**14**	**271**	**38**	**1996, 1997**	Schweigert and Callomon, 1997	Schweigert and Callomon, 1997
Crussol, composite of 8 sections	**Valence, France**	44°56'N	4°51'E	**Eudoxus - Planula**	**9**	**146**	**174**	**1995, 1996**	Atrops 1982	Atrops 1982; Hantzpergue et al., 1997
Composite of 6 sections	**Iberian Range, Spain**	41°04'N	1°00'W	**Bimammatum - Bifurcatus**	**49**	**147**	**408**	**1993, 1994**	Goy et al., 1981; Steiner et al., 1986	Bello 1994; Bello et al., 1995; Meléndez 1989; Meléndez and Fontana, 1991
						Total =	**940**			

were sampled and analyzed (by M.W. Hounslow; and by J.G. Ogg with A.L. Coe); and each group exchanged and re-interpreted the datasets, thereby giving us confidence in the combined paleomagnetic interpretations.

The South Ferriby Clay Pit in North Lincolnshire is an important reference section for the Oxfordian–Kimmeridgian boundary in the Sub-Boreal faunal province (e.g. Cox, 2001; Wright, 2003, see Table 1; Supplementary Fig. 2). The dark claystone strata were deposited on the East Midland shelf, adjacent to the North Sea Basin system (Rawson and Wright, 1995).

The Ringstead Bay and nearby Black Head sections (Supplementary Figs. 3 and 4) are part of the extensive Late Jurassic exposures on the Dorset coast. The sedimentology and interpreted sequence stratigraphy are summarized by Coe (1992, 1995), and the ammonite zonation is based on Wright (1986, 1998, 2001, 2003). The classical Oxfordian–Kimmeridgian boundary section at Ringstead Bay is marked by a stratigraphic gap and regrettably now buried under a coastal protection scheme, which eliminated this as a candidate for the international GSSP. We were fortunate to have taken extensive samples prior to its burial, and the magnetic polarity results from the *Ringsteadia pseudocordata*, *Pictonia baylei* and *Rasenia cymodoce* Sub-Boreal Zones are important for inter-regional correlation above and below the Oxfordian–Kimmeridgian boundary.

2.2. Sub-Mediterranean sections from Western Europe

We incorporate three studies of portions of the Upper Oxfordian and Lower Kimmeridgian that have Sub-Mediterranean ammonite species.

A suite of sections at Aguilón, Tosos and Moneva, NE Iberian Range, Spain, confirmed a polarity pattern for the *Perisphinctes luciaeformis* Subzone (*Gregoryceras transversarium* Zone) through the *Euaspidoceras hypselum* Subzone (*Epipeltoceras bimammatum* Zone). The Oxfordian–Kimmeridgian stratigraphy of the NE Iberian Range is summarized by Meléndez (1989) (see also Table 1). The earlier magnetostratigraphic composite by Steiner et al. (1986) was replicated and significantly expanded with additional sections by Juárez et al. (1994) and Juárez et al. (1995). Unfortunately, details for the biostratigraphy (ammonite ranges, zonal definitions, etc.) have yet to be published for any of these excellent magnetostratigraphic sections; and the differing placement of zonal boundaries relative to polarity zones among these three publications has not been elucidated. We have incorporated the multi-section synthesis in the composite section (Supplementary Fig. 13).

A section at Plettenberg in the Swabian Alb (Germany), spanning the lower *Idoceras planula*–upper *Epipeltoceras bimammatum* zones had been a potential candidate for the base-Kimmeridgian GSSP (Schweigert and Callomon, 1997). We re-interpreted the demagnetization behavior of all samples reported in an earlier magnetostratigraphic study (Erik Settles, unpublished Ph.D. Thesis, Ludwig-Maximilian-Universität München, 1999) to derive a polarity succession and paleomagnetic pole (Supplementary Fig. 11).

The classic ammonite-rich limestone strata of Crussol (near Valence, France), yielded a magnetostratigraphy from multiple quarries and hillside exposures that spanned the uppermost Oxfordian (sensu Sub-Mediterranean), entire Kimmeridgian and lowermost Tithonian (Atrops, 1982; Hantzpergue et al., 1997). In this paper, we incorporate the succession from the uppermost Oxfordian (*Sutneria galar* Subzone of the *Idoceras planula* Zone) to the uppermost Kimmeridgian (*Hybonoticeras beckeri* Zone) (Supplementary Fig. 12).

2.3. Sub-Mediterranean sections from Poland

Outcrops and quarries in strata in Poland are generally condensed in the lowermost Middle and Lower Oxfordian or span only one or two ammonite zones, therefore had yielded an incomplete pattern of magnetozones (Kądziałko-Hofmokl and Kruczyk, 1987; Ogg et al., 1991; Ogg and Gutowski, 1996). We undertook an extensive two-summer campaign to study over twenty stratigraphic sections from this region to obtain and verify a complete Oxfordian and Lower Kimmeridgian magnetostratigraphy. Seven of these sections encompass portions from the uppermost *Perisphinctes bifurcatus* Zone (upper *Perisphinctes grossouvrei* Subzone) to the middle *Idoceras planula* Zone (upper *Idoceras planula* Subzone) (Fig. 2; Table 1). Detailed bio- and lithostratigraphic studies of most of these sections were published by Matyja and Wierzbowski (see Table 1). Additional biostratigraphic levels obtained by this team during the 2007 field season are incorporated into the magnetostratigraphy (Supplementary Figs. 5 through 10).

3. Analysis of paleomagnetic samples

3.1. Demagnetization procedure

The suites of paleomagnetic samples from the British, Polish, French and German sections were analyzed in five laboratories (University of Michigan in USA, Lancaster University and Oxford University in Britain, GeoForschungsZentrum Potsdam and the University of Munich in Germany, and Polish Academy of Sciences). Remanent magnetization was measured using two- or three-axis cryogenic magnetometers.

The majority of samples from the Tethyan Realm (Poland, France and Germany) are limestones and clayey limestones. The samples from the Boreal Realm (Britain) are dominated by silty to calcareous claystones (South Ferriby, Isle of Skye, upper portion of Ringstead Bay). Fissile clay-rich minicores of British samples were generally wrapped in aluminum foil to prevent disintegration during analysis. Such wrappings had no measurable magnetization.

We tailored the demagnetization procedure for each lithology of each section according to pilot studies of progressive thermal and alternating-field treatments. In general, the demagnetization for most samples consisted of 25–30 °C steps within the typical range that yielded characteristic directions. Samples were heated stepwise to a maximum of 600 °C. Thermal steps were duplicated for samples with weaker intensities. Magnetic susceptibility was generally monitored after every second thermal demagnetization step below 300 °C and after every temperature step above 300 °C. On the 2G magnetometer, the alternating-field (AF) demagnetization was performed along three axes from zero to maximum 120mT, usually in steps of every 5mT. A sub-set of specimens from the Staffin section were analyzed using stepwise thermal treatments up to 240 °C, then additional steps were obtained using a Molspin tumbling AF demagnetizer.

Stratigraphic details, progressive demagnetization data and sets of interpretations for each sample of each outcrop are provided in the Supplementary Data Appendix of this report.

3.2. Interpretation of magnetic behavior and magnetic carriers

The clay-rich lithologies from the British sections display a rapid decrease in magnetic intensity upon heating to either 80 °C or 150 °C, during which the directions tended to converge toward a present-day field direction. In such claystones, potentially goethite may be a major carrier of NRM (e.g., France and Oldfield, 2000). Divergence to a second, much weaker component of dual-polarity direction, typically occurred above 200° to 250 °C. However, above about 350 °C, most samples displayed a rapid increase in magnetic susceptibility which may be the combined effect of iron-rich clay dehydration and pyrite oxidation and decomposition (e.g., Ellwood et al., 1986; Cairanne et al., 2004). Similar trends were exhibited by the samples from sideritic-limestone concretions. Characteristic directions were generally obtained from the clay-rich samples between 250°–330 °C.

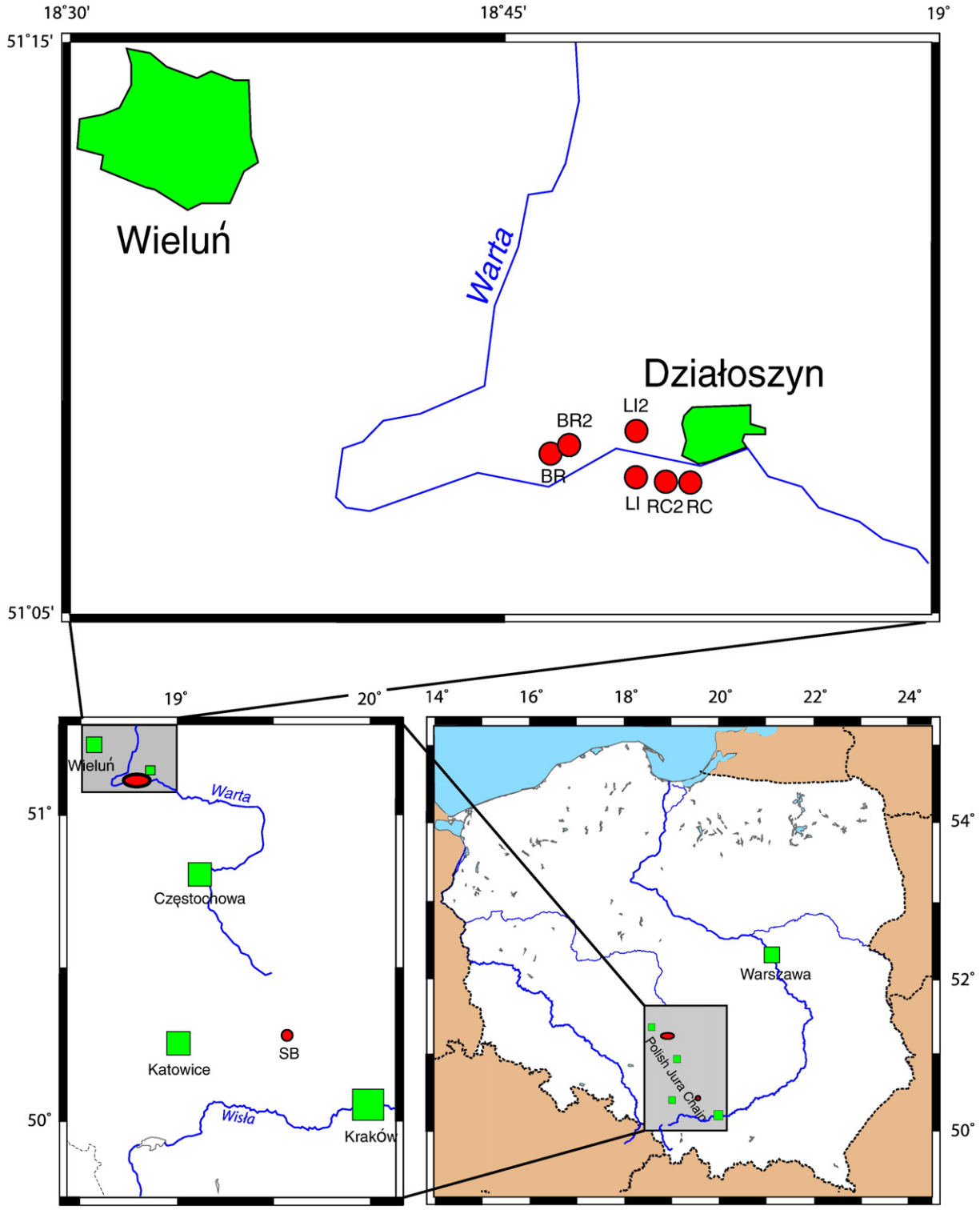


Fig. 2. Location of Polish sections in Sub-Mediterranean faunal province. Outcrops, marked as red circles from Wieluń to Kraków, are: Bobrowniki (BR and BR2), Lisowice (LI = PJ110), Lisowice2 (LI2 = PJ139–140), Raciszyn (RC = PJ113–114), Raciszyn2 (RC2 = PJ125–126), and Syborowa Góra (SB); see also outcrop abbreviations in Matyja and Wierzbowski (1997).

Additional details of the magnetic behavior of claystones from Flodigarry are summarized in Wierzbowski et al. (2006).

In general, limestones, which dominated the suites from the Sub-Mediterranean faunal province, had much lower magnetic susceptibilities and enabled a better resolving of characteristic direction vectors than the claystones. Low-temperature steps progressively removed normal-polarity-type components. Divergence into dual-polarity directions was achieved above 300 °C. Susceptibility remained stable through 500 °C, although by that step, most samples displayed a magnetization near the noise level of the magnetometers (approximately below 5.0×10^{-6} A/m). Characteristic directions were generally interpreted within the 200°–400 °C interval. The typical

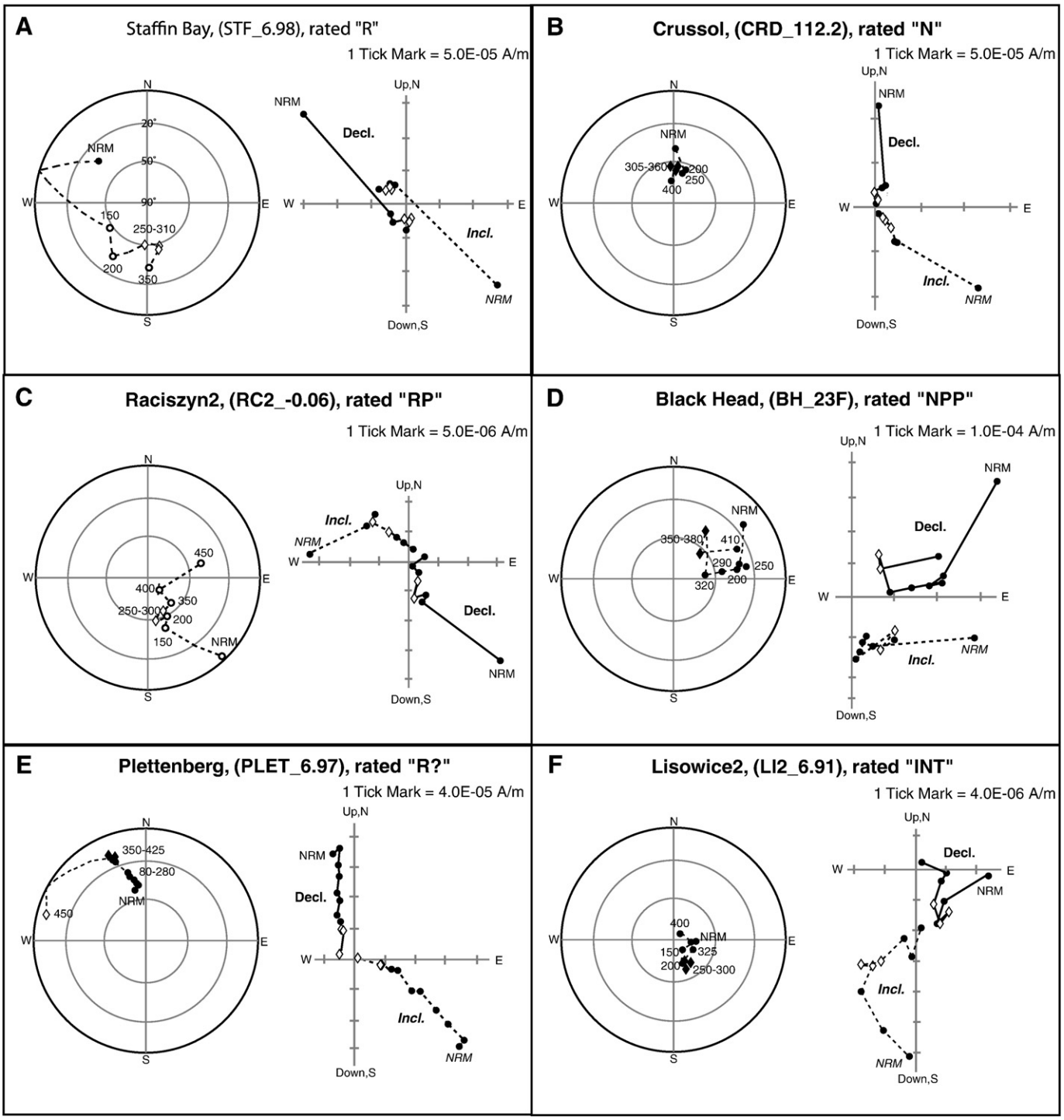


Fig. 3. Examples of demagnetization plots for different lithologies, polarities and quality ratings. Behavior of selected samples of each assigned quality rating during progressive thermal demagnetization is displayed as stereographic projection (left side of each pair) and projected vector (Zijderveld) diagram (right side). Declination and inclination vectors in the Zijderveld diagrams (solid lines versus dashed lines) correspond to the horizontal component and to the actual dip of the magnetization vector; steps incorporated in computation of the characteristic direction are indicated by diamonds. (A) Staffin Bay STF_6.98 (calcareous limestone) from the Isle of Skye has a reversed polarity rated "R" based upon its clustering. Characteristic vector obtained from 3 thermal steps, 250°–310° = 173.7° declination, −47.8° inclination, 3.4×10^{-5} A/m intensity. (B) Crussol CRD_111.2 (light-gray micritic limestone) has normal polarity. Characteristic vector (3 thermal steps, 305°–360° = 0.0° declination, 54.3° inclination, 5.0×10^{-5} A/m intensity) is rated "N" based upon its tight clustering and uni-vectorial decay. (C) Raciszyn2 RC2_−0.06 (light-gray micritic limestone) is rated "RP" based upon its clear reversed-polarity cluster and stable uni-vectorial decay but weak intensity. Characteristic vector is from only 2 measurements (250°–300° = 167.6° declination, −49.7° inclination, 7.6×10^{-6} A/m). Intensity at 350 °C and higher was near the noise level of the magnetometer. (D) Black Head BH_23F (grayish brown sandstone) is rated "NPP" based on its unstable normal trend and only two characteristic steps. Obvious normal polarity, but not useful for computing the mean poles. Characteristic vector is from 2 thermal steps (350°–380° = 38.0° declination, 44.5° inclination, 1.6×10^{-4} A/m). (E) Plettenberg PLET_6.97 (thin-bedded limestone) is rated "R?" based upon its migrating trend towards reverse polarity with apparent flip to reversed-polarity hemisphere at the final 450° thermal step. Perhaps reversed polarity, but uncertain. This sample is located at the transitional zone from normal to reversed polarity. (F) Lisowice LI2_6.91 (chalky limestone) is rated "INT" (indeterminate) due to its unequivocal polarity characterized by reversed declination and positive inclination typical for normal samples. Characteristic vector is from 3 measurements (250°–300° = 149.9° declination, 61.6° inclination, 1.4×10^{-5} A/m).

magnetic behavior during thermal demagnetization suggests that a common carrier of characteristic magnetization is magnetite, with a possible contribution from magnetic sulfides.

3.3. Interpretation of polarity and calculation of paleomagnetic directions

Most samples either attained a direction associated with normal or reversed polarity which remained stable through multiple heating steps as the intensity of magnetization decreased (i.e., uni-vectorial decay to the origin), or displayed development of a semi-stable direction prior to the onset of erratic magnetization at higher temperatures. Some samples did not attain a clear characteristic component, but displayed great circle paths towards either of the dual-polarity directions. This was especially true for many of the mudstones from the Isle of Skye before onset of the thermally-induced surge in susceptibility and acquisition of spurious magnetic behavior. Examples of the range of magnetic behaviors and associated interpretation (and uncertainty) of polarity are shown in Fig. 3.

Characteristic directions of magnetization were computed using the least-squares method of Kirschvink (1980). Some characteristic directions were computed by the technique of great circle analysis (e.g., reversed-polarity samples of Plettenberg by Eric Settles, unpublished Ph.D. thesis (1999); and several sampling horizons in the Isle of Skye section by Mark Hounslow, reported in Wierzbowski et al., 2006).

Characteristic directions were assigned a polarity and reliability rating. If the directions from at least three successive demagnetization steps were less than 15° from each other and had converged toward the main normal- or reversed-polarity clustering, then that specimen was assigned a reliable normal- (N) or reversed-polarity (R) ranking. The NP or RP rankings were assigned to specimens in which the successive steps of demagnetization were displaced by more than 15° from each other, or if the characteristic direction of the specimen was derived from only two steps. Less precise directions, but with a definite trend toward normal or reversed-polarity trend, were indicated as NPP or RPP. Uncertain polarity was given a N? or R? rank if a great circle trend was interpreted. Specimens with ambiguous, indeterminate or intermediate polarity were rated as INT. The magnetic behavior, characteristic direction, polarity assignment and quality ranking of each specimen were analyzed independently at least by two observers (Ogg, Przybylski; plus Hounslow for Isle of Skye), and a compromise assignment was chosen after discussion.

4. Paleomagnetic poles

4.1. Calculation of paleomagnetic directions and paleopoles

Specimens rated as N, R, NP and RP were used to calculate mean directions, paleomagnetic poles and paleolatitudes, with their associated uncertainties. Mean directions were calculated separately for normal-, reversed- and combined-polarity directions using a weighted Fisher mean (Fisher, 1953) in which the NP and RP-rated specimens were given half-weight. In a second iteration, outliers of more than two circular standard deviations from the mean were excluded. For comparison, an additional mean direction was computed by adding the mean vectors from normal- and reversed-polarity samples. In some cases where the average intensities of polarity intervals are similar, this vector sum can yield a less-distorted mean characteristic direction, because a pervasive secondary magnetic vector that affects equally the samples of each polarity will be cancelled during this addition of polarity vectors. In addition, the vector addition method may reduce the distortion if there is a preponderance of samples with one polarity in a section, because it gives equal weight to each specimen.

The pair of normal- and reversed-polarity poles was subjected to two significance tests. The reversal test of Butler (1998) is positive if the confidence limit of the mean normal direction falls inside the confidence limit of the inverted mean direction of the reverse polarity. If these confidence limits do not overlap but the combined mean direction is contained within the 5% significance level (with 95% confidence), an additional significance test is required. This test is positive if the calculated Fisher coefficient F is smaller than its tabulated value for $p=0.05$ for 2 and 2(N-2) degrees of freedom (Ft). The reversal test of McFadden and McElhinny (1990) evaluates a critical angle (γc) between two Fisher mean directions at a particular confidence level (5% level used here). If γc is larger than the angle between the mean normal and inverted reverse directions (γ), the test indicates the directions are drawn from the same population (i.e. a positive test). A positive reversal test is further classified as 'A' if $\gamma c \leq 5°$, as 'B' if $5° < \gamma c \leq 10°$, as 'C' if $10° \leq \gamma c \leq 20°$ and as 'indeterminate' if $\gamma c > 20°$. Unfortunately, the majority of the sections contained too few high quality-rated samples (N, R) for each polarity (e.g., the McFadden and McElhinny reversal test requires more than five samples for each of the polarities). Therefore, regional data composites were prepared from the characteristic directions of the British, Polish and the French sections (Supplementary Fig. 1). This technique also allowed an overview of all paleomagnetic directions and eliminated outliers — specimens with stable but anomalous directions from the main clusters, which may indicate persistent secondary

Notes to Table 2
Abbreviations
Number, N: total number of observations (specimens) used in the analysis (fractional if data is weighted)
Decl: mean geographic declination
Incl: mean geographic inclination
Intensity: mean intensity of the ChRM directions
α95 [°]: angular radius of the cone of 95% confidence level
K: Fisher precision parameter
R: length of the resultant vector of all individual directions
Lat, Long latitude and Longitude of the virtual geomagnetic paleopole
δm: radius of the 95% confidence level ellipse of the paleomagnetic pole, measured along the direction of the magnetic meridian from the site to the vertical pole
δp: radius of the 95% confidence level ellipse of the paleomagnetic pole, measured perpendicular to the direction of the magnetic meridian from the site to the vertical pole
Paleo Lat: magnetic paleolatitude of the sampling location
+/− on Paleo Lat: upper and lower confidence limits on the magnetic paleolatitude; mean values of these limits are given
F: Fisher statistic for reversal test (Butler, 1998)
Ft: tabulated F value for $p=0.05$ for 2 and 2(N-2) degrees of freedom
γ: spherical angle between the mean normal and mean inverted reverse direction
γ_c: reversal critical angle (McFadden and McElhinny, 1990)
Sites: number of studies (for Europe from APWP) or localities
Oxf.–Kim. bound.: Oxfordian–Kimmeridgian boundary
Mid. Oxf.: Middle Oxfordian
APWP: Apparent Pole Wander Path

Table 2
(A) Mean directions and paleopoles for Upper Oxfordian–Lower Kimmeridgian, and significance tests of data sets, (B) datasets not included in calculation of the mean paleopoles due to rotated tectonic blocks, and (C) comparison of mean directions, pole position and paleolatitudes with previous studies.

A. Mean directions and paleopoles for Upper Oxfordian - Lower Kimmeridgian and significance tests of data sets

		After bedding correction													Reversal tests	
Location	Magnetic Directions	Least squares (weighted)							Mean virtual geomagnetic pole						Butler 1998	McFadden and
		Number	Decl [°]	Incl [°]	Intensity [A/m]	α95 [°]	*k*	*r* / *n*	Lat [°]	Long [°]	δp[°]	δm[°]	Paleo Lat [°]	upper (+)	lower (-)	F / Ft	McElhinny, 1990 Y / Yc
British Composite (Ringstead Bay, Black Head and South Ferriby)	Normal	13.5	1.3	52.0	2.84E-04	8.3	25.0	13.00/13.5	70.6	175.0	7.7	11.3	32.6			**0.31/3.26**	**5.39/13.00**
	Reverse	6.0	172.8	−53.8	4.27E-04	7.1	89.9	5.94/6.0	−71.6	17.3	6.9	9.9	−34.3			Positive test	Positive test, class C
	Combined	19.5	358.7	52.6	3.22E-04	5.9	32.3	18.93/19.5	**71.2**	**181.6**	5.6	8.2	**33.2**	39.3	27.9		
Polish Jura Chain, Poland	Normal	24.5	6.3	56.0	2.21E-05	4.0	56.0	24.08/24.5	74.9	179.1	4.1	5.7	36.6			**3.05/3.10**	**5.10/5.41**
	Reverse	22.0	186.4	−61.1	1.60E-05	3.8	66.9	21.69/22.0	−80.2	−9.9	4.5	5.9	−42.2			Positive test	Positive test, class B
	Combined	46.5	6.3	58.5	1.90E-05	2.8	58.4	45.72/46.5	**77.4**	**175.9**	3.0	4.1	**39.2**	42.3	36.2		
Plettenberg, Swabian Alb, Germany	Normal	19.0	0.0	50.5	2.36E-05	5.1	44.2	18.59/19.0	70.1	187.9	4.6	6.9	31.3			**0.43/3.32**	**5.55/15.07**
	Reverse	2.5	182.6	−55.9	1.29E-05	26.4	37.4	2.46/2.5	−75.1	−0.4	27.1	37.8	−36.4			Positive test	Test not possible -
	Combined	21.5	0.3	51.1	2.20E-05	4.7	44.8	21.04/21.5	70.6	187.2	4.4	6.4	31.8	36.4	27.7		not enough Reverse
	Vector comb.		0.9	52.4	1.82E-05				**71.8**	**185.6**	4.5	6.5	**33.0**				directions
Crussol, Valence, France (CRA - CRF)	Normal	37.0	5.2	52.1	3.04E-05	2.6	83.7	36.57/37.0	77.1	164.9	2.4	3.5	32.7			**2.59/3.10**	**5.42/5.74**
	Reverse	9.0	177.9	−49.2	3.36E-05	5.2	100.9	9.92/9.0	−75.1	11.8	4.5	6.8	−30.1				
	Combined	46.0	3.7	51.5	3.10E-05	2.3	83.1	45.46/46.0	77.0	170.9	2.2	3.2	32.2	34.4	30.1	Positive test	Positive test, class B
	Vector comb.		1.3	50.6	3.19E-05				**76.4**	**180.2**	2.1	3.1	**31.3**				
Combined	Based on the mean position of paleopoles	**4.0**				**2.7**	**598.5**	**3.995/4.0**	**74.2**	**181.3**	**3.8**	**3.8**					

B. Data not Included in Calculation of the Mean Paleopole

		After Bedding Correction												References
Location	Magnetic directions	Least-squares					Mean virtual geomagnetic pole							
		Number	Decl [°]	Incl [°]	α95 [°]	*k*	Lat [°]	Long [°]	δp[°]	δm[°]	Paleo Lat [°]	upper (+)	lower (-)	
Isle of Skye, UK	Great circle Mean	48.0	17.0	45.0	6.6	10.7	**57.0**	**146.0**	5.3	8.4	26.6	32.2	21.6	Present study
Aguilón and Tosos, Iberian Range, Spain	Normal	139.0	324.9	42.3	2.6	22.4	56.5	250.3	1.97	3.20				Juárez et al., 1994
	Reverse	129.0	143.6	−38.0	5.2	6.6	−53.5	67.5	3.63	6.15				
	Combined	268.0	324.1	40.6	2.9	9.9	**55.9**	**251.2**	2.12	3.5	**23.2**	25.4	21.1	
Aguilón, Iberian Range, Spain	Combined	96.0	322.1	44.5	3.1	22.0	**55.2**	**251.8**	2.4	3.8	**26.2**	**28.7**	23.8	Steiner et al., 1986
Moneva, Iberian Range, Spain	Combined	83.0	330	48	5	12	**63**	**257**	4	7	**29**	34	25	Juárez et al., 1995

C. Comparison with previous studies

Location	Stratigraphic age	Sites	Decl [°]	Incl [°]	α95 [°]	*k*	Lat [°]	Long [°]	Paleo Lat [°]	References
European Craton	Oxf.-Kim. bound.	14	5.8	55.2	2.7	598.5	**74.2**	**181.3**	35.8	Present study
Polish Jura Chain	Oxf.-Kim. bound.	5	6.3	58.5	2.8	58.4	**77.4**	**175.9**	39.2	
Polish Jura Chain	Callovian-Mid.Oxf.	4	359.6	54.8	3.6	665.0	**74.3**	**200.3**	35.3	Ogg et al., 1991
Polish Jura Chain	Callovian-Mid.Oxf.	6	16.8	57.2	7.3	58.0	**72.3**	**150.4**	37.8	Kądziako-Hofmonkl and Kruczyk, 1987
Europe from APWP	Lower Tithonian (mean age)	10	12.0	58.0	6.6	54.3	**75.0**	**159.9**	38.6	Besse and Courtillot, 2002
Europe from APWP	Upper Bathonian (mean age)	15	18.5	58.5	5.0	59.7	**72.5**	**144.0**	39.2	

overprints. Regional normal- and reversed-polarity mean directions and paleopoles were computed from these composite datasets, and tested with the statistical methods (Table 2). In order to compare coeval directions, the Crussol composite direction is only for the interval of the uppermost Oxfordian (*Sutneria galar* Subzone of the *Idoceras planula* Zone) through the Lower Kimmeridgian (*Crussoliceras divisum* Zone), rather than including the upward extension through the full Kimmeridgian and lowermost Tithonian.

4.2. Mean pole for cratonic Europe, and regional tectonics of Iberia and Isle of Skye

All four paleopoles from the British, Polish and French composites and the Plettenberg section of Germany are within ~6° of each other. Therefore, merging these four different regions on the European Craton yields a very precise pole for the Late Oxfordian through Early Kimmeridgian at 74.2°N, 181.3°E (A_{95} = 3.8°) (Table 2A).

The magnetic directions and apparent pole from the Flodigarry section on the Isle of Skye in Scotland are anomalous. First, the Isle of Skye site yields a paleolatitude of 26.6°N, in contrast to the expected paleolatitude of ~40°N. We suggest that inclination shallowing following setting of magnetization during the burial compaction of the claystone-dominated sediments is a contributing factor. Second, the Isle of Skye section displays similar rotation trends (~20° clockwise) regardless of site bedding corrections, which suggests that this represents a post-Jurassic clockwise rotation of this section relative to the European Craton. This local rotation may be an artifact associated with the large-scale landslide which contains the section.

The paleolatitudes of the array of sites on the European Craton (excluding the Isle of Skye) ranged from about 39°N (Poland) to 31°N (southern France). This is similar to the paleolatitudes derived from the previous studies of the Upper Bathonian–Callovian and Oxfordian (Table 2C). All paleolatitudes shown in this Table 2C were calculated for the mean location of the Polish Jura outcrops (51.0°N, 18.9°E), based on the mean virtual paleopoles taken from the literature (references in Table 2C). Mean paleopoles for the Upper Bathonian and Lower Tithonian derived from the averaged continental apparent polar wander paths by Besse and Courtillot (2002) are rotated clockwise about 11° and 6° relative to our pole at the Oxfordian–Kimmeridgian boundary. Even though the upper Middle and Late Jurassic (150–160 Ma) portion of the global analysis by Besse and Courtillot (2002) is based on only a few poles relative to any other interval in the 200 myr span of their compilation, their analysis suggests a rapid rate of polar wander relative for Europe during this interval. Our compilation provides a pole based on many sites for a narrow time interval during the postulated broad episode of rapid continental motion.

The Iberian region underwent a major counterclockwise rotation during the opening of the Bay of Biscay along with other post-Oxfordian tectonic motions (e.g., Steiner et al., 1986; Juárez et al., 1994, 1995). Therefore, even though the ~26°N paleolatitudes of the suite of Spanish sites during the Middle Oxfordian are consistent with the projected position of Iberia (e.g., Satolli et al., 2007, their Fig. 8), these sites imply a total post-Oxfordian counterclockwise rotation of about 60° relative to the cratonic Europe.

5. Magnetic polarity composite for ammonite zonations

5.1. Boreal and Sub-Boreal composite

An integrated stratigraphy for each section was compiled with lithostratigraphy, magnetostratigraphy, selected sequence stratigraphy, and ammonite datums and associated zones (Fig. 4, Supplementary Figs. 2 through 4). The polarity pattern relative to ammonite subzones is generally quite consistent (Fig. 5). The uppermost Oxfordian is dominated by normal polarity. In contrast, the lowermost Kimmeridgian (sensu Boreal) is predominantly reversed polarity with two or three thin normal-polarity subzones. The major change in polarity dominance is very close to the Oxfordian–Kimmeridgian boundary (i.e., *Pictonia flodigarriensis* horizon of the *Pictonia densicostata* Sub-Boreal Subzone). The exact level for this major change in polarity dominance relative to the proposed GSSP has a slight uncertainty due to sampling limitations. The majority of the *Pictonia flodigarriensis* horizon in the main inter-tidal section at Skye consists of an approximately 1-m interval of soft, fissile claystone that is typically a seawater-filled depression and was not sampled for magnetostratigraphy by either of the two independent studies due to the difficulty in obtaining coherent samples. Future sampling of this horizon would be desirable to obtain a more precise inter-regional paleomagnetic correlation at the Oxfordian–Kimmeridgian Stage boundary. However, for most purposes of inter-regional correlation, the proposed global Oxfordian–Kimmeridgian boundary essentially coincides with the polarity reversal.

5.2. Sub-Mediterranean composite

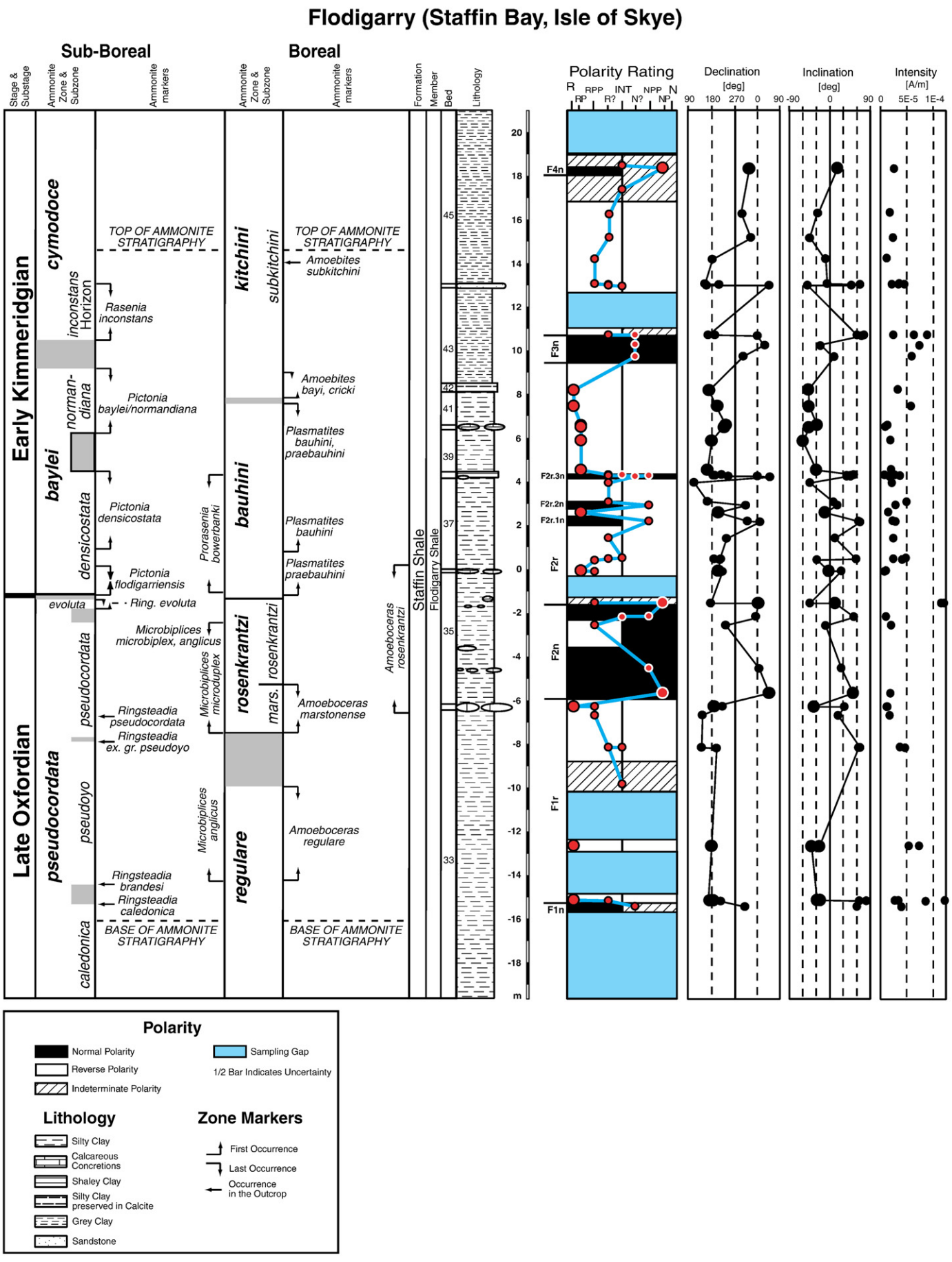
Normal polarity dominates in the lower part of the Plettenberg quarry, which corresponds to the *Epipeltoceras bimammatum* Subzone of the *Epipeltoceras bimammatum* Zone. The overlying Beds 12–21, attributed to the *tizianiformis* horizon of the *Taramelliceras hauffianum* Subzone (cf. Schweigert and Callomon, 1997), show a marked change from normal polarity (below) to reversed polarity (above) at the base of Bed 17. However, the biostratigraphic interpretation of the *Epipeltoceras bimammatum–Taramelliceras hauffianum* subzonal transition in the Plettenberg section (Beds 12–19) is not unequivocally proven. The paleomagnetic data suggest that the bulk of the interval in question may represent the *Epipeltoceras bimammatum* Subzone (Supplementary Fig. 11).

Younger parts of the Polish sections from the Upper Oxfordian (sensu Sub-Mediterranean; i.e. the lowermost Kimmeridgian sensu Boreal) and the corresponding upper part of the Plettenberg section show a dominance of reversed polarity with relatively thin intervals of normal polarity. This polarity pattern is typical for the upper part of the *Epipeltoceras bimammatum* Subzone, the entire *Taramelliceras hauffianum* Subzone, and the bulk of the *Idoceras planula* Zone (Fig. 5; and Supplementary Figs. 5 through 12). Based on the position of the Polish sections with respect to ammonite subzones, their array of the reversed-dominated polarity is projected to be above the extensive sections of the NE Iberian Range (Fig. 6, Supplementary Fig. 13).

In contrast, the topmost part of the *Idoceras planula* Zone in the Mt. Crussol section (the *Sutneria galar* Subzone, and the overlying deposits of the *Sutneria platynota* and *Ataxioceras hypselocyclum* zones) is dominated by normal polarity.

The *Upper Amoeboceras Layer* present in Polish sections at Raciszyn and Raciszyn2 contains both Boreal and Sub-Boreal species (e.g. *Amoeboceras bauhini* and *Pictonia densicostata*). The *bauhini* horizon at Plettenberg is an important reference level as the type horizon of *Amoeboceras bauhini* (Oppel); it is much thicker than the *Upper Amoeboceras Layer* from the Polish sections, and shows a more detailed paleomagnetic pattern (cf. Supplementary Figs. 11, 7 and 8). Therefore, the *bauhini* horizon of Plettenberg probably represents a longer time interval than the *Upper Amoeboceras Layer* from the Polish sections. The biostratigraphic comparison of the Plettenberg

Fig. 4. Magnetostratigraphy of the Flodigarry section, Staffin Bay, Isle of Skye, which is the candidate for the Kimmeridgian GSSP. Stratigraphic column and biostratigraphy are modified from Matyja et al. (2006). Polarity rating: N/R — good normal- or reversed-polarity behavior, INT — indeterminate or intermediate. The larger dots indicate directions of samples used in the paleomagnetic analysis. The intensity column displays the value of the computed vector for the characteristic direction of each sample.


Flodigarry (Staffin Bay, Isle of Skye)
Sub-Boreal
Boreal
Stage & Substage
Ammonite Zone & Subzone
Ammonite markers
Formation
Member
Bed
Lithology
Polarity Rating
Declination [deg]
Inclination [deg]
Intensity [A/m]
Early Kimmeridgian
Late Oxfordian
cymodoce
baylei
pseudocordata
caledonica
inconstans Horizon
norman-diana
densicostata
evoluta
pseudocordata
pseudoyo
kitchini
subkitchini
bauhini
rosenkrantzi
mars.
regulare
TOP OF AMMONITE STRATIGRAPHY
BASE OF AMMONITE STRATIGRAPHY
Rasenia inconstans
Pictonia baylei/normandiana
Pictonia densicostata
Pictonia flodigarriensis
Ring. evoluta
Microbiplices microbiplex, anglicus
Ringsteadia pseudocordata
Ringsteadia ex. gr. pseudoyo
Ringsteadia brandesi
Ringsteadia caledonica
Microbiplices microduplex
Microbiplices anglicus
Prorasenia bowerbanki
Amoebites subkitchini
Amoebites bayi, cricki
Plasmatites bauhini, praebauhini
Plasmatites bauhini
Plasmatites praebauhini
Amoeboceras rosenkrantzi
Amoeboceras marstonense
Amoeboceras regulare
Staffin Shale
Flodigarry Shale
F4n
F3n
F2r.3n
F2r.2n
F2r.1n
F2r
F2n
F1r
F1n
Polarity
Normal Polarity
Reverse Polarity
Indeterminate Polarity
Sampling Gap
1/2 Bar Indicates Uncertainty
Lithology
Silty Clay
Calcareous Concretions
Shaley Clay
Silty Clay preserved in Calcite
Grey Clay
Sandstone
Zone Markers
First Occurrence
Last Occurrence
Occurrence in the Outcrop

succession with the Polish succession indicates a possible stratigraphic gap at Plettenberg above the *bauhini* horizon, which encompasses the youngest part of the *Taramelliceras hauffianum* Subzone, and the oldest part of the *Idoceras planula* Zone (cf. Schweigert and Callomon, 1997; Matyja and Wierzbowski, 1997).

The Sub-Mediterranean polarity composite (Fig. 6) has a small gap in sampling-coverage in the mid-parts of the *Epipeltoceras bimammatum* Zone at the boundary between the *Euaspidoceras hypselum* and *Epipeltoceras bimammatum* Subzones. As explained later, the coeval magnetostratigraphic sections in Britain indicate that this interval corresponds to a zone of predominantly normal polarity in the Boreal and Sub-Boreal realm.

5.3. Sub-Mediterranean to Boreal magnetostratigraphy correlation and composite

The correlations between the British (Boreal) composite and the Western Europe (Tethyan) composite are constrained by rare incursions of Boreal- and Sub-Boreal ammonite species into the Sub-Mediterranean Province (e.g. *Amoeboceras* ammonite-bearing layers in the Polish Jura Chain and SW Germany shown in detailed biostratigraphic studies, e.g., Matyja and Wierzbowski, 1992, 1997, 2002, 2006a,b; Głowniak, 2006). These constraints on the correlation among faunal provinces near the Oxfordian–Kimmeridgian boundary imply that much of the upper *Epipeltoceras bimammatum* Zone and *Idoceras planula* Zone (historically considered Upper Oxfordian) of the Sub-Mediterranean Province should be coeval with the lowest Kimmeridgian (*Amoeboceras bauhini* Zone and *Amoeboceras bayi* horizon of the *Amoebites kitchini* Zone) of the Boreal Province (Matyja et al., 2006). Therefore, the magnetozone dominated by reversed polarity of the lowest Kimmeridgian in the British sections (BB3) corresponds to the reversed-polarity dominance obtained from the Polish suite of Sub-Mediterranean outcrops (SM3) (see Figs. 5–7 and comments above in Section 5.2).

The major upward shift from normal to reversed polarity is recorded in Staffin Bay near the boundary between the uppermost *Amoeboceras rosenkrantzi* and the lowermost *Amoeboceras bauhini* Boreal ammonite zones, close to the proposed Kimmeridgian–Oxfordian boundary (between magnetozones BB3 and BB4). This normal-to-reversed-polarity transition below the Boreal base of the Kimmeridgian corresponds in the Sub-Mediterranean succession to the boundary between magnetozones SM3 and SM4 in the *Epipeltoceras bimammatum* Subzone in Polish outcrops at Bobrowniki and probably at Raciszyn, and in the German quarry at Plettenberg (Fig. 6; see also Supplementary Figs. 8, 9 and 11).

In the Boreal Realm, the underlying normal-polarity zone extends downward to the middle *Amoeboceras marstonense* Subzone (the middle of magnetozone BB6, Figs. 5 and 7). In the Sub-Boreal Province, this normal-polarity interval at South Ferriby and Black Head sections extends from the uppermost *Ringsteadia evoluta* downward to the lower *Ringsteadia pseudocordata* ammonite subzone. In the Sub-Mediterranean Realm, the lower part of this major normal-polarity interval is recorded in the *Euaspidoceras hypselum* Subzone at Syborowa Góra quarry and in the Spanish composite sections (the middle of magnetozone SM5, Figs. 6 and 7).

The normal-polarity zone in the Syborowa Góra succession contains the *Lower Amoeboceras Layer* with *Amoeboceras ovale* and *Epipeltoceras semimammatum* ammonites, implying a correlation to the lower *Euaspidoceras hypselum* Sub-Mediterranean Subzone. The interval of reversed polarity between the upper *Amoeboceras regulare* and lower *Amoeboceras rosenkrantzi* Boreal zones is constrained to correlate with the Sub-Mediterranean succession from the upper *Perisphinctes grossouvrei* and lower *Euaspidoceras hypselum* Subzones (Fig. 7, magnetozones BB6 and SM5, accordingly).

From this set of first-order correlations, we propose a set of more detailed correlations of individual "shorter" normal-polarity subchrons within the main reversed-polarity interval (the Boreal BB2–BB3 magnetozones, corresponding to the Sub-Mediterranean magnetozones SM2 and SM3), plus matching of the succession of polarity zones below the normal-dominated polarity at the Boreal magnetozones BB6–BB7 and the Sub-Mediterranean magnetozones SM5–SM6 (Fig. 7). There remain minor problems caused by the uncertainties in equating ammonite subzones due to inadequately delimited taxa ranges (e.g. due to the rarity of Boreal/Sub-Boreal ammonites in the Sub-Mediterranean Province), relatively poorly known succession of Sub-Mediterranean ammonites in the upper *Euaspidoceras hypselum*–lowermost *Taramelliceras hauffianum* stratigraphic interval, the presence of some intervals with indeterminate polarity or possible pervasive overprints (e.g., lowermost Crussol section, magnetozone SM2), and the minor incompleteness of British and Sub-Mediterranean polarity patterns (magnetozones BB3, BB7 and SM4). However, the main features are quite consistent.

We merge the patterns from these two faunal provinces into a master composite that fills the gaps in one region by the proposed correlative pattern in the other region (Fig. 7). The only two remaining uncertain intervals in this merged composite are: (1) the questionable normal-polarity-dominated (overprinted?) zone in the lowermost Crussol section which is apparently temporally correlative with a reversed-polarity-dominated zone in the Boreal and Sub-Boreal realm at magnetozone BB2 (this interval remains to be verified through replicate magnetostratigraphic sections in the Sub-Mediterranean Province that span the middle part of the *Idoceras planula* Zone), and (2) the polarity of the interval corresponding to the bulk of the *Pictonia flodigarriensis* horizon at the base of the *Pictonia baylei* Zone in the Skye section (it covers an ~1-meter-thick interval between the normal-polarity interval of *Ringsteadia pseudocordata* Zone and the overlying reversed-polarity interval of *Pictonia baylei* Zone).

5.4. Correlation to M-sequence marine magnetic anomalies

The merged polarity pattern is generally consistent with the features of marine magnetic anomalies M24B through M28 as obtained from geomagnetic surveys from the Western Pacific and Eastern Indian Ocean (e.g., Sager et al., 1992; 1998; Tivey et al., 2006; Tominaga et al., 2008) (Fig. 8).

The details of the marine anomaly clusters called M25A and M26 are poorly resolved due to the very weak amplitude of the magnetic signal recorded in the currently available study of that part of the Pacific Ocean (Handschumacher et al., 1988). We re-interpreted the relative widths of polarity blocks comprising these M25A and M26 anomaly clusters based on survey data from the Eastern Indian Ocean (Sager et al., 1992). Our re-interpretation is generally consistent with the merged polarity composite. Our outcrop-based magnetostratigraphy resolved 2 of the 3 modeled subchrons of M25A and all four subchrons of M26 (see Fig. 8).

Subchron M25A-a was not resolved in the Sub-Mediterranean Realm, probably due to the overprint in the lowermost part of the Crussol section, nor has it been confirmed in the British sections. Reversed-polarity Chron M27r is possibly indicated by a single-sample horizon at South Ferriby (Fig. 5, Supplementary Fig. 2). Additional magnetostratigraphic sections, denser sampling or marine magnetic surveys are required to adequately resolve the fine details of the magnetic pattern within these intervals. However, the majority of the marine magnetic polarity anomalies can be correlated to the outcrop magnetostratigraphy, hence to the ammonite zonations.

The Oxfordian–Kimmeridgian boundary of the Boreal Realm, which will be the future global GSSP definition, is close to the boundary between Chron M27n and Chron M26r. The marine magnetic anomalies can be used to calibrate the relative durations of ammonite biozones and chrons of geomagnetic polarity time scale (Supplementary Fig. 14). When a robust numerical age interpolation for this portion of the M-Sequence is obtained from radiometric ages,

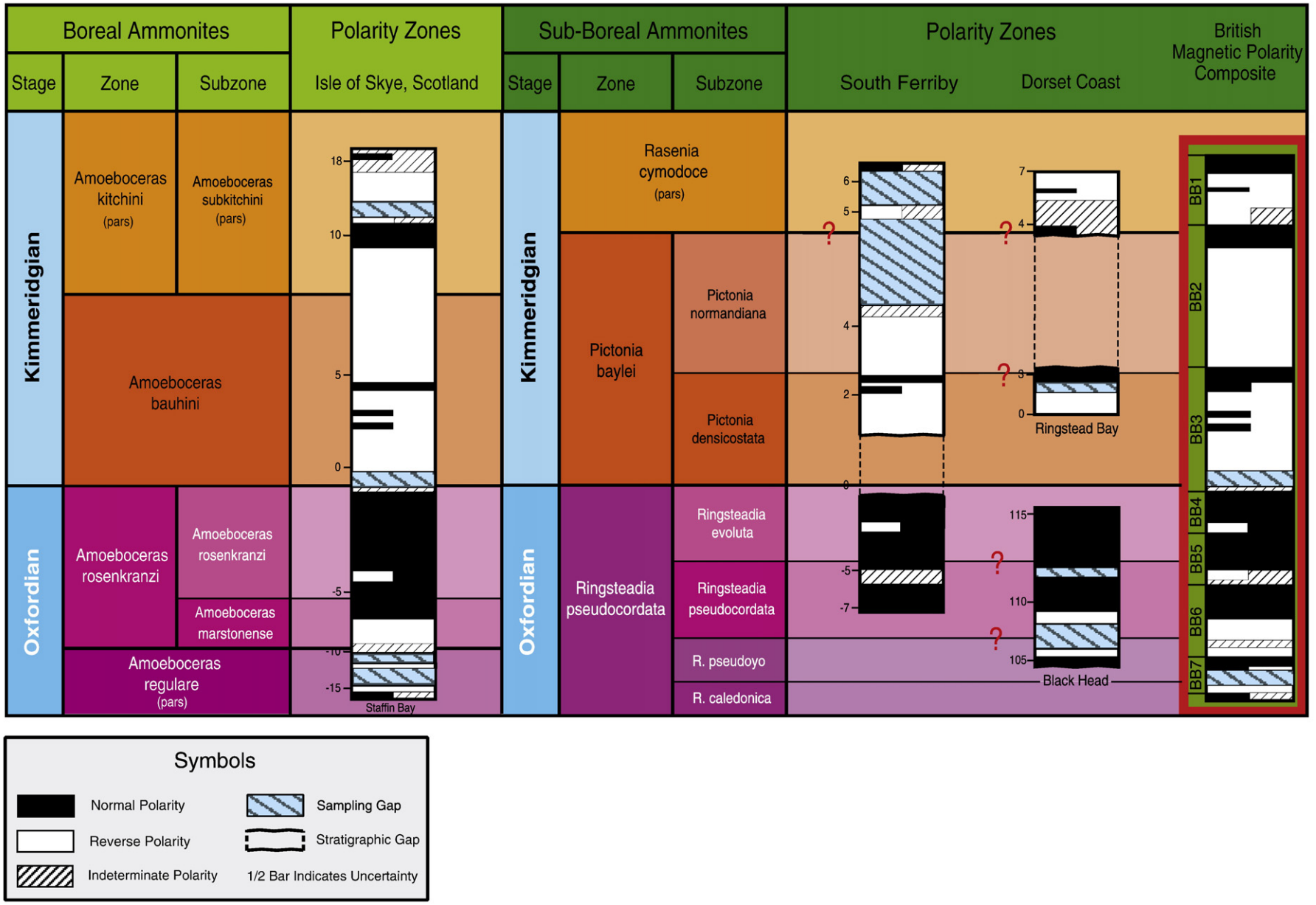


Fig. 5. Magnetostratigraphic sections and composite polarity pattern of the Oxfordian–Kimmeridgian boundary interval in the British region of the Boreal and Sub-Boreal faunal provinces.

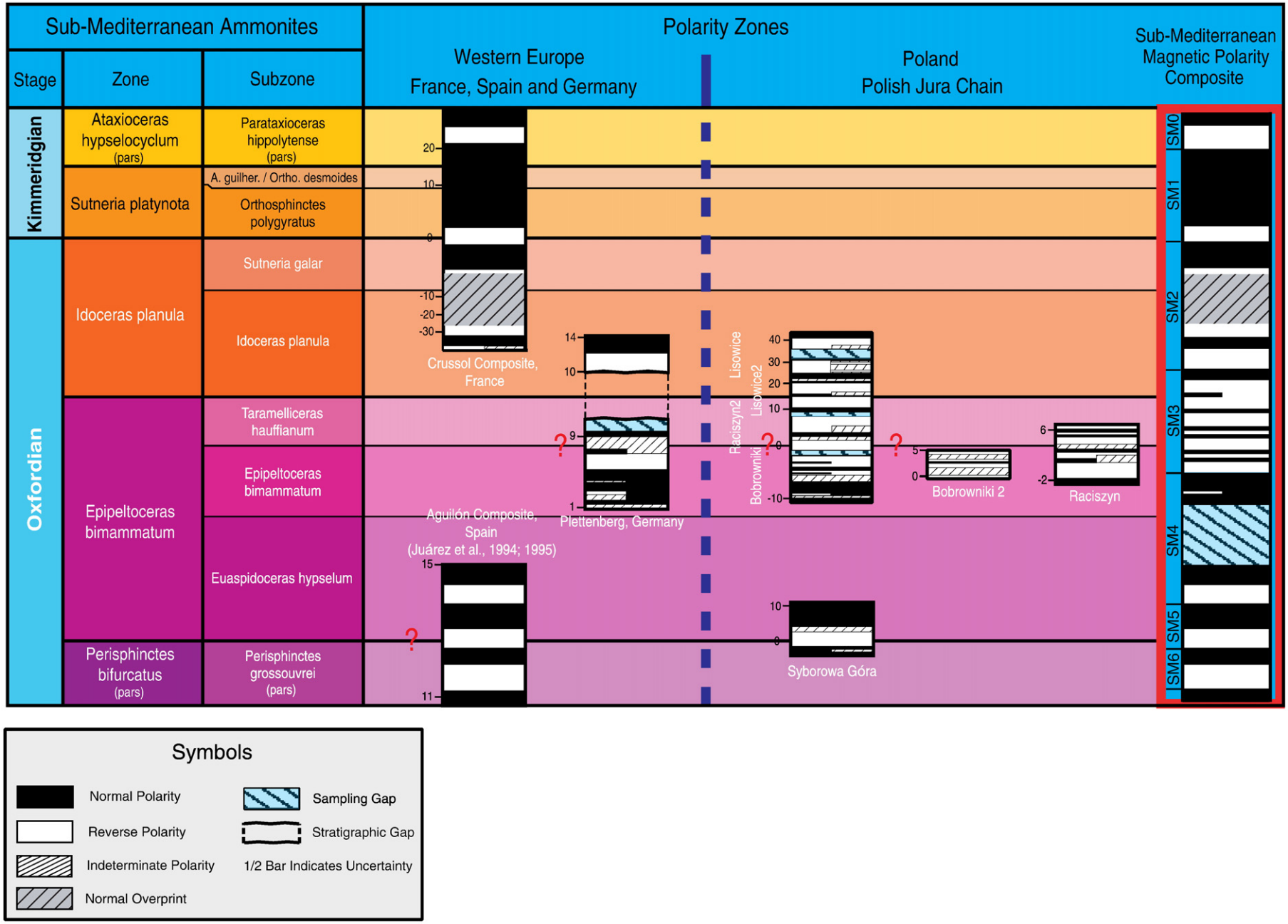


Fig. 6. Magnetostratigraphic sections and composite polarity pattern of the Oxfordian–Kimmeridgian boundary interval in the Sub-Mediterranean faunal province.

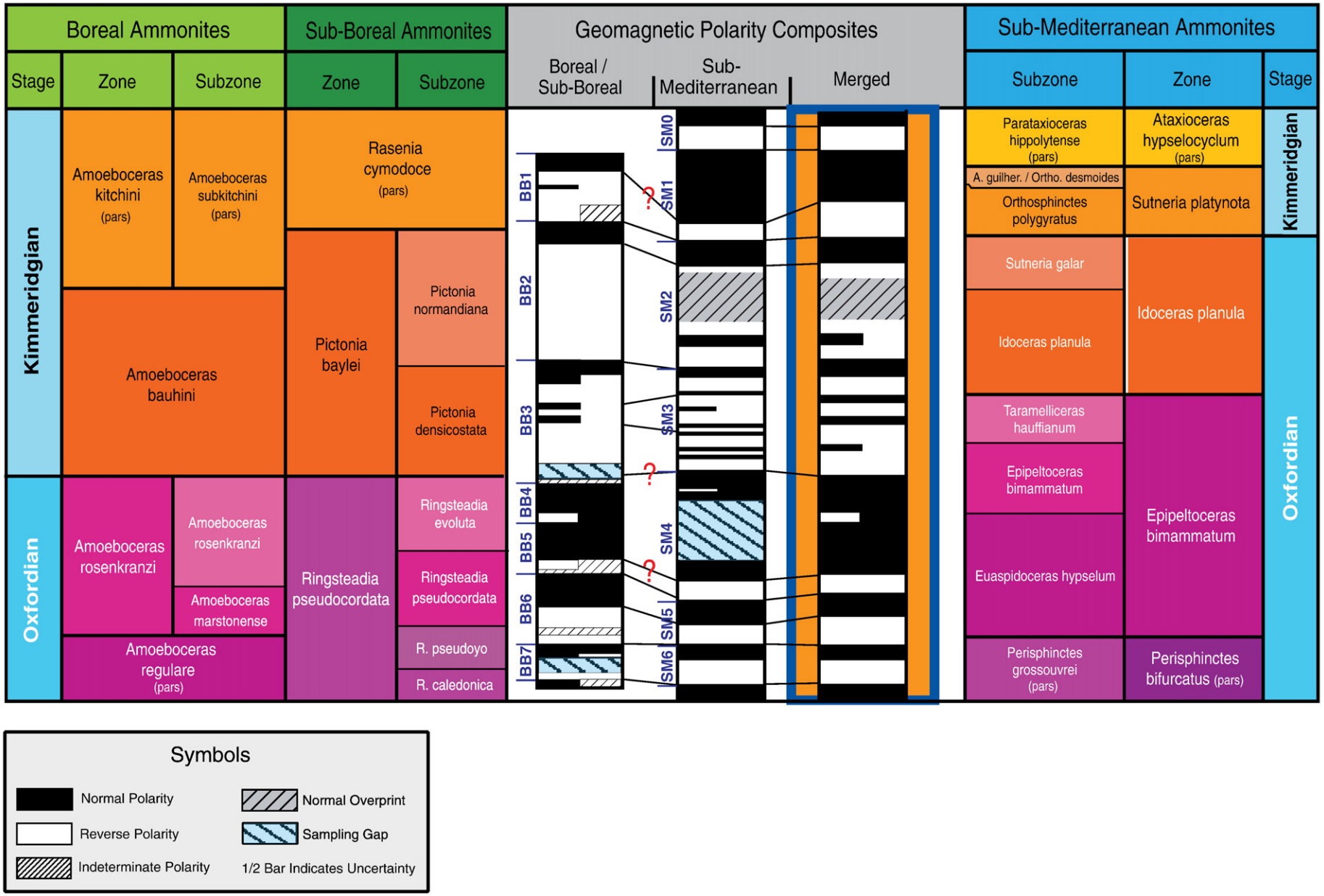


Fig. 7. Correlation of the geomagnetic polarity composite between faunal realms.

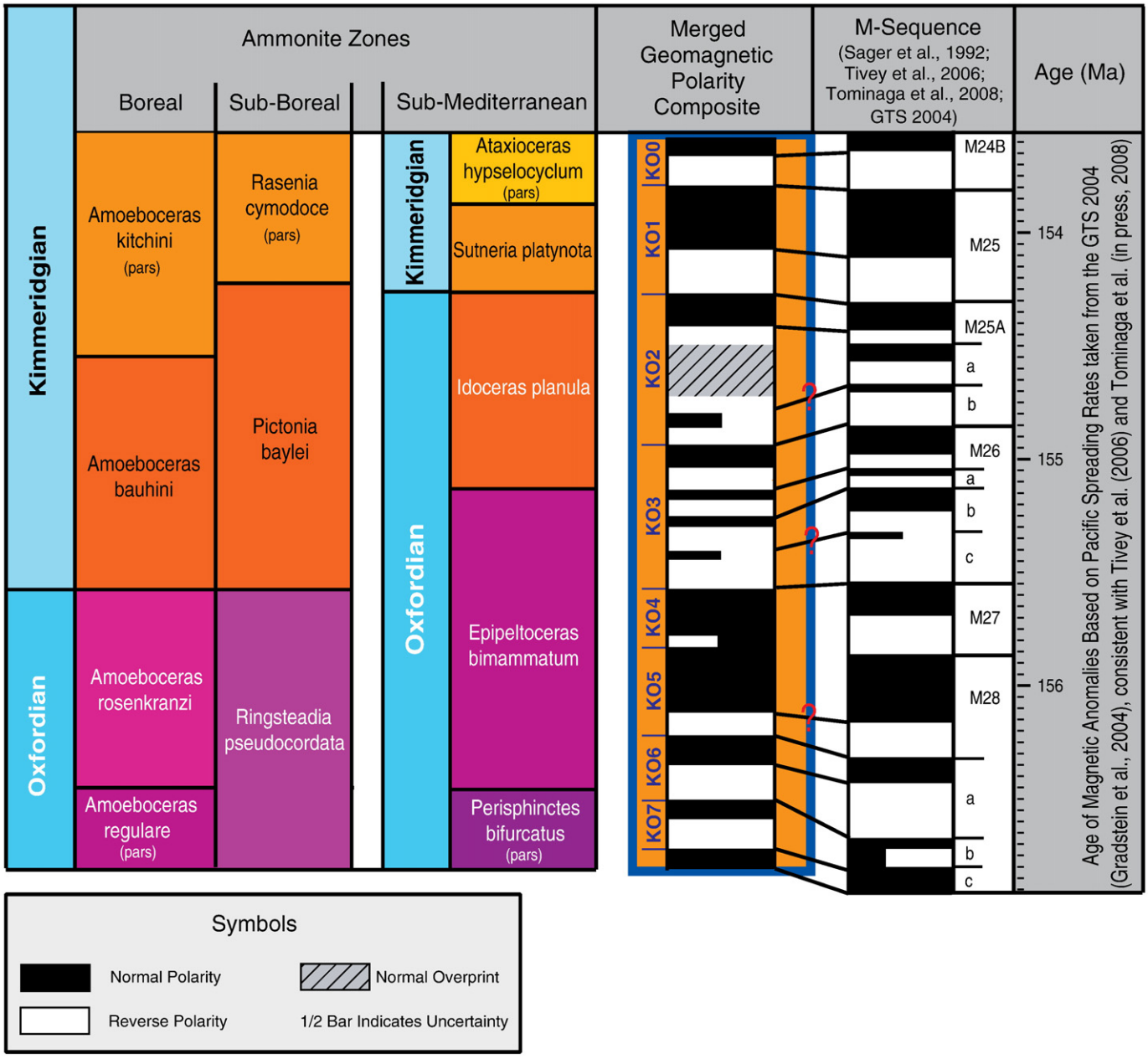


Fig. 8. Correlation of the geomagnetic polarity composite with the M-sequence of marine magnetic anomalies.

cycle stratigraphy and oceanic spreading models, then it will be possible to assign ages to these biozones.

6. Conclusions

The combined biostratigraphy and magnetostratigraphic results from sections in the Boreal, Sub-Boreal and Sub-Mediterranean outcrops allow a confident correlation of the Oxfordian–Kimmeridgian Stage boundary interval (as defined in Staffin Bay of the Isle of Skye) among these faunal provinces. The base of the Kimmeridgian, assigned in the GSSP as the base of the *Amoeboceras bauhini* and *Pictonia baylei* Zones, lies very close to the base of a reversed-polarity dominated interval. A similar polarity signature obtained from Sub-Mediterranean outcrops from Poland and Germany and marine magnetic-anomaly records indicates that this level approximately corresponds to the lower to middle part of the *Epipeltoceras bimammatum* Subzone and to the base of Chron M26r. This implies that the Oxfordian–Kimmeridgian Stage boundary of the Boreal faunal province is approximately 1.2 myr older than the traditional Sub-Mediterranean version of Oxfordian–Kimmeridgian Stage boundary (base of *Sutneria platynota* Zone).

The paleopole calculated for the Oxfordian–Kimmeridgian boundary from the combined data sets of Britain, Poland, France and Germany is 74.2°N, 181.3°E ($A_{95}=3.8°$). This pole is consistent with previous compilations; although a direct comparison is difficult the other studies did not examine this limited time interval or had derived a synthetic mean that averaged multiple stages.

The composite polarity pattern enables a calibration of ammonite subzones to marine magnetic anomalies M24b through M28 and an improved polarity model for the poorly-resolved details of the magnetic-anomaly clusters comprising M25A and M26. Assigning a constant spreading rate for this portion of the M-sequence enables an estimation of the relative durations for the biozones within each faunal province. Improvements in numerical dating of the M-sequence or in the cycle stratigraphy of Oxfordian–Kimmeridgian

biozones (e.g., Boulila et al., 2008) will lead to an improved model of seafloor spreading rates and age assignments for the Oxfordian and Kimmeridgian.

For global correlation purposes, the Oxfordian–Kimmeridgian boundary is close to the base of Chron M26r.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.epsl.2009.11.014.

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